

Koopman-Based Online Identification With Sim2Real Transfer for Hydrodynamic Modeling of Turtle Inspired Robot

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Abstract—This letter addresses critical challenges in the dynamic modeling of biomimetic underwater robots, including heavy reliance on prior knowledge and difficulties in tracking time-varying models. To tackle these issues, we propose a control-oriented online identification framework based on the Koopman operator and successfully apply it to the hydrodynamics modeling of a sea turtle-inspired biomimetic robot. Initially, a prior model is obtained offline through simulation data based on Koopman theory. Subsequently, an online update algorithm integrated with Kalman filtering is developed to calibrate the model parameters in real time using a small amount of experimental data. Simulation and experimental results demonstrate that the proposed Koopman model outperforms baseline methods in long-term prediction accuracy and generalization capability. After online updates, the model achieves a 42.13% improvement in prediction accuracy over the baseline. In orientation tracking experiments under flow disturbances, the proposed framework enhances control accuracy by 63.4% compared to the baseline, while exhibiting smaller overshoot and stronger disturbance rejection capabilities. Overall, this study provides valuable insights for high precision dynamic modeling of biomimetic underwater robots.

Index Terms—Online system identification, bionic turtle robot, Koopman operator, Kalman filter.

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I. INTRODUCTION

BIO-INSPIRED underwater robots (BURs), leveraging their exceptional biological compatibility [1] and low environmental disturbance characteristics [2], have become indispensable tools in fields such as marine environmental monitoring [3] and archaeological exploration [4]. Compared to traditional propeller driven autonomous underwater vehicles (AUVs) and remotely operated vehicles (ROVs), BURs generate significantly lower noise [5], hydrodynamic wake [6], and sediment resuspension during locomotion [7]. For instance, Katzschnmann et al. successfully utilized acoustic communication to control a robotic fish for close observation of marine life, providing an effective platform for studying organisms in their natural habitats [8].

However, the unique morphologies [9], numerous control degrees of freedom [10], and diverse bio-inspired propulsion mechanisms of these robots [11] pose significant challenges for precise dynamic modeling. To address this, researchers have proposed various methods [12]. Wang et al. established a three dimensional motion model for a robotic fish, including helical motion, based on Euler's equations of motion and grey box model hydrodynamic estimation [13]. Nonetheless, these methods typically rely on physical prior knowledge, including specific robot configurations, and determining model parameters often requires extensive time for meticulous design and experimental calibration. In previous work [14], although the single rigid body dynamic equations were employed to describe the motion characteristics of the amphibious turtle robot, the model still struggles to adequately capture the dynamic features arising from the three dimensional oscillation of the flippers.

Fortunately, data driven approaches, such as deep learning [15], sparse identification of nonlinear dynamics (SINDy) [16], and the Koopman operator [17], offer promising solutions for identifying nonlinear system behaviors solely from motion data. Deep learning involves solving nonlinear optimization problems and may suffer from convergence issues [15], while SINDy requires pre-constructing a dictionary of candidate functions [16], which can be restrictive for BURs with limited prior knowledge. In contrast, the Koopman operator framework globally linearizes nonlinear systems by lifting them into an infinite dimensional linear space while preserving control information. Koopman based system identification has been successfully applied to various systems, including spherical robots [18], soft continuum manipulators [19], and wheeled robots [20]. Notably, Pan et al. proposed a Bayesian Koopman framework for automatic

基于Koopman的在线识别及Sim2Real迁移用于受乌龟启发的机器人水动力建模

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Abstract——本文针对仿生水下机器人动态建模中的关键挑战, 包括对先验知识的高度依赖以及难以跟踪时变模型的问题。为解决这些问题, 我们提出了一种基于Koopman算子的面向控制的在线识别框架, 并成功将其应用于海龟仿生机器人水动力学建模。首先, 通过基于Koopman理论的仿真数据离线获得先验模型。随后, 开发了一种与卡尔曼滤波集成的在线更新算法, 利用少量实验数据实时校准模型参数。仿真和实验结果表明, 所提出的Koopman模型在长期预测精度和泛化能力上均优于基线方法。在线更新后, 该模型在预测精度上比基线提高了42.13%。在方向跟踪实验中,

Index Terms——在线系统识别、生物仿真龟机器人、库普曼算子、卡尔曼滤波器。

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一、引言

仿生水下机器人(BURs)利用其卓越的生物相容性[1]和低环境干扰特性[2], 已成为海洋环境监测[3]和考古勘探[4]等领域不可或缺的工具。与传统的螺旋桨驱动自主水下航行器(AUVs)和遥控水下航行器(ROVs)相比, BURs在运动过程中产生的噪声[5]、水动力尾迹[6]和沉积物再悬浮[7]显著更低。例如, Katzschmann等人成功利用声学通信控制机器人鱼以近距离观察海洋生物, 为在自然栖息地研究生物提供了有效平台[8]。

然而, 这些机器人的独特形态[9]、众多控制自由度[10]以及多样的仿生推进机制[11]对精确的动态建模提出了重大挑战。为了解决这一问题, 研究人员提出了各种方法[12]。王等人建立了一种用于机器人鱼的三维运动模型, 包括基于欧拉运动方程和灰箱模型水动力估计的螺旋运动[13]。尽管如此, 这些方法通常依赖于物理先验知识, 包括特定的机器人配置, 并且确定模型参数通常需要大量时间进行细致的设计和实验校准。在以前的工作[14]中, 尽管采用了单刚体动力学方程来描述两栖乌龟机器人的运动特性, 但该模型仍然难以充分捕捉由鳍肢三维振动产生的动态特征。

幸运的是, 数据驱动的方法, 例如深度学习[15]、非线性动力学稀疏识别(SINDy)[16]和库普曼算子[17], 为仅通过运动数据识别非线性系统行为提供了有前景的解决方案。深度学习涉及求解非线性优化问题, 可能存在收敛性问题[15], 而SINDy需要预先构建候选函数字典[16], 这对于先验知识有限的BURs来说可能存在限制。相比之下, 库普曼算子框架通过将非线性系统提升到无限维线性空间, 同时保留控制信息, 从而实现非线性系统的全局线性化。基于库普曼的系统识别已成功应用于各种系统, 包括球形机器人[18]、软体连续操纵器[19]和轮式机器人[20]。值得注意的是, Pan等人提出了一个用于自动化的贝叶斯库普曼框架

controller generation, effectively reducing the manual burden of constructing lifting functions, as validated on robotic fish platforms [21].

Despite reducing reliance on physical prior knowledge, Koopman-based methods still require substantial datasets for training [22]. Acquiring sufficient stable motion data for BURs is often challenging due to experimental constraints. A feasible strategy is to use dynamic simulators to generate diverse motion data for prior model identification, which is then updated and corrected with a small amount of real-world experimental data, enabling a Simulation-to-Reality (Sim2Real) transfer. This strategy has been widely adopted in embodied AI, including legged and humanoid robotics [23]. For example, the fine grained medium model proposed by Choi et al. enabled a reinforcement learning agent to achieve good adaptation to unfamiliar environments, overcoming limitations of the training data distribution [24].

Furthermore, BURs face the challenge of time varying dynamics in practical applications: factors such as varying payloads [18], mechanical wear of components [13], and coupling among multiple controllers [12] can lead to drift in dynamic parameters. Therefore, developing robust identification algorithms capable of online tracking of model variations is crucial. For instance, addressing the online identification of soft environmental dynamic models, Schindeler et al. proposed a two stage polynomial linearized Hunt-Crossley model identification method, achieving high precision online identification of nonlinear contact dynamics via a look up table recursive least squares estimator [25].

In summary, this study first identifies a Koopman model for the bio-inspired turtle robot based on simulation data. Subsequently, online updating of the model parameters is achieved using experimental data and a Kalman filter. The validity of the proposed framework is further validated through figure-eight path tracking simulations and directional tracking experiments. The main contributions are as follows:

- A Koopman operator modeling approach tailored for control tasks is proposed. The method significantly outperforms baseline methods in long-term prediction accuracy and generalization capability across different reference signals and control tasks. In simulations, the control accuracy is improved by 59.6% compared to baseline methods.
- An online update mechanism incorporating Kalman filtering is designed and implemented. This enhances the model's ability to track time-varying dynamics. On experimental datasets, the prediction accuracy is improved by 42.13% compared to baseline methods.
- Experimental validation of directional tracking is conducted in a pool with flow disturbances. The proposed framework achieves a 63.4% improvement in control accuracy over baseline methods, along with smaller overshoot and stronger disturbance rejection capability.

The letter is organized as follows: Section II introduces the preliminaries. Section III elaborates on the system identification algorithm based on Koopman theory. Section IV outlines the process of data driven identification and subsequent validation. Section V details the experimental implementation of an online Kalman filter based update mechanism for the Koopman model. Finally, Section VI presents the conclusions.

II. PRELIMINARIES

This section presents the design and mathematical modeling of the turtle inspired robot.

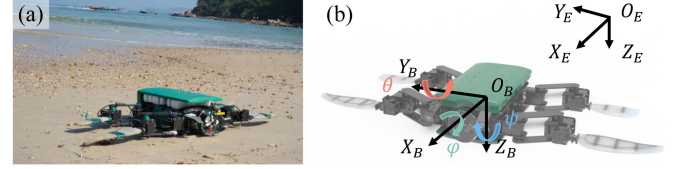


Fig. 1. Prototype and coordinate system. (a) The prototype of bio-inspired turtle robot. (b) An established definition of coordinate systems for the amphibious robotic platform, specifying the body fixed frame and the global inertial reference frame.

TABLE I
TECHNICAL SPECIFICATIONS OF THE BIO-INSPIRED TURTLE ROBOT

Item	Value & Description
Dimensions	$0.43 \times 1.23 \times 0.17 \text{ m}^3$
Weight	10.48 kg
Actuation Mechanism	3-DOF flippers
Sensing Modules	Camera, Pressure Sensors, Power Meter
Computing Unit	Raspberry Pi 4B (4 GB)
IMU	MPU-9250, 100 Hz sampling frequency
Servo Motors	GXYipin servo with 130 kg-cm torque
Endurance time	2 h
UWB	DW1000, accuracy error $\leq 100\text{mm}$

A. Design of the Turtle Inspired Robot

Turtles in nature, with their flattened body structure and powerful flippers, exhibit both excellent locomotor stability and efficient propulsion capabilities. Based on these biological features, we have designed a turtle inspired robot, whose prototype is shown in Fig. 1(a), with key structural parameters listed in Table I. The robot primarily consists of a central hull and a pair of symmetrically arranged three degree of freedom bionic flippers.

B. Mathematical Modeling

Fig. 1(b) illustrates the coordinate system for the turtle-inspired robot. The dynamic equation of the turtle inspired robot can be expressed as follows:

$$M(q)\dot{\mu} + C(\mu)\mu + D(\mu)\mu + g(q) = J^T F + S_a \tau \quad (1)$$

where q denotes the generalized coordinate and μ denotes the generalized velocities. The matrices $M(q)$, $C(\mu)$, and $D(\mu)$ represent the system inertia matrix, coriolis and centrifugal matrix, and hydrodynamic damping matrix, respectively. The vectors $g(q)$, F , and τ denote the gravitational and buoyancy forces/moments, external forces/moments, and generalized torque input, respectively. J represents Jacobian matrix toward applying points of F . The selective matrix S_a denotes the operations performed on τ .

Considering that the robot exhibits bilateral symmetry in both the longitudinal lateral (x - y) and longitudinal vertical (x - z) planes, the coupled motions among surge, sway, and yaw can be neglected. During locomotion, the turtle utilizes its forelimbs as the primary propulsion mechanism while the hindlimbs serve for directional control. Based on this biomimetic principle and engineering simplification objectives, and referring to recent study [26], we determine that only surge and yaw require active control in planar motion. Consequently, (1) can be simplified to a horizontal plane parametric hydrodynamic model that incorporates added mass, as well as linear and nonlinear damping

控制器生成,有效减少了构建提升函数的人工负担,在机器人鱼平台上已得到验证 [21]。

尽管减少了对物理先验知识的依赖,基于 Koopman 的方法仍然需要大量数据进行训练 [22]。由于实验条件的限制,获取足够稳定的 BUR 运动数据通常具有挑战性。一种可行的策略是使用动态模拟器生成多样化的运动数据以进行先验模型识别,然后用少量真实实验数据进行更新和校正,从而实现从仿真到现实 (Sim2Real) 的迁移。这一策略已在具身 AI 中被广泛采用,包括腿式和人形机器人 [23]。例如,Choi 等人提出的细粒度中等模型使强化学习代理能够很好地适应不熟悉的环境,克服了训练数据分布的限制 [24]。

此外,BUR在实际应用中面临时间变化动力学的挑战:诸如变化的有效载荷 [18]、组件的机械磨损 [13] 以及多个控制器之间的耦合 [12] 等因素都可能导致动力学参数的漂移。因此,开发能够在线跟踪模型变化的鲁棒识别算法至关重要。例如,为了应对软环境动力学模型的在线识别,Schindeler 等提出了一种两阶段多项式线性化 Hunt-Crossley 模型识别方法,通过查找表递归最小二乘估计器实现高精度的非线性接触动力学在线识别 [25]。

总之,本研究首先基于仿真数据为仿生乌龟机器人识别了一个Koopman模型。随后,利用实验数据和卡尔曼滤波实现了模型参数的在线更新。通过八字形路径跟踪仿真和方向跟踪实验,进一步验证了所提出框架的有效性。主要贡献如下:

- 提出了一种针对控制任务的 Koopman 算子建模方法。该方法在长期预测精度和跨不同参考信号及控制任务的泛化能力上明显优于基线方法。在仿真中,与基线方法相比,控制精度提高了 59.6%。设计并实现了一种结合卡尔曼滤波的在线更新机制,这增强了模型跟踪时变动态的能力。在实验数据集上,与基线方法相比,预测精度提高了 42.13%。
- 在有流动干扰的水池中进行方向跟踪的实验验证。所提出的框架在控制精度上比基线方法提高了 63.4%,同时具有更小的超调和更强的抗干扰能力。

本文结构如下:第 II 节介绍了预备知识。第 III 节阐述了基于 Koopman 理论的系统辨识算法。第 IV 节概述了数据驱动的辨识过程及后续验证。第 V 节详细说明了基于在线卡尔曼滤波的 Koopman 模型更新机制的实验实现。最后,第 VI 节给出了结论。

II. 初步工作

本节介绍了仿乌龟机器人的设计和数学建模。

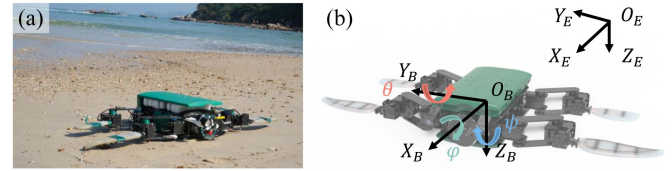


图 1. 原型及坐标系。(a) 仿生乌龟机器人的原型。(b) 为两栖机器人平台建立的坐标系定义,指定了机体固定坐标系和全局惯性参考坐标系。

表 I 仿生海龟机器人技术规格

Item	Value & Description
Dimensions	$0.43 \times 1.23 \times 0.17 \text{ m}^3$
Weight	10.48 kg
Actuation Mechanism	3-DOF flippers
Sensing Modules	Camera, Pressure Sensors, Power Meter
Computing Unit	Raspberry Pi 4B (4 GB)
IMU	MPU-9250, 100 Hz sampling frequency
Servo Motors	GXYipin servo with 130 kg·cm torque
Endurance time	2 h
UWB	DW1000, accuracy error $\leq 100\text{mm}$

A. Design of the Turtle Inspired Robot

自然界中的乌龟,凭借其扁平的身体结构和强壮的鳍足,表现出卓越的运动稳定性和高效的推进能力。基于这些生物特征,我们设计了一种受乌龟启发的机器人,其原型如图1(a)所示,主要结构参数列于表I中。该机器人主要由一个中央船体和一对对称布置的三自由度仿生鳍足组成。

B. Mathematical Modeling

图1(b)说明了受乌龟启发的机器人的坐标系统。受乌龟启发的机器人的动力学方程可以表示如下:

$$M(q)\dot{\mu} + C(\mu)\mu + D(\mu)\mu + g(q) = J^T F + S_a \tau \quad (1)$$

其中 q 表示广义坐标, μ 表示广义速度。矩阵 $M(q)$ 、 $C(\mu)$ 和 $D(\mu)$ 分别表示系统惯性矩阵、科氏力和离心力矩阵以及水动力阻尼矩阵。向量 $g(q)$ 、 F 和 τ 分别表示重力和浮力/力矩、外部力/力矩以及广义力矩输入。 J 表示作用点 F 的雅可比矩阵。选择矩阵 S_a 表示对 τ 执行的操作。

考虑到该机器人在纵向横向 (x - y) 和平面纵向垂直 (x - z) 上表现出双侧对称,因此可以忽略前后、横向和平移的耦合运动。在运动过程中,海龟利用其前肢作为主要推进机制,而后肢用于方向控制。基于这一仿生原理和工程简化目标,并参照近期研究 [26],我们确定在平面运动中仅需对前后运动和偏航进行主动控制。因此,(1) 可以简化为一个水平面参数化水动力模型,该模型包含附加质量以及线性和非线性阻尼。

effects:

$$\begin{aligned}\dot{x} &= u \cos \psi - v \sin \psi \\ \dot{y} &= u \sin \psi + v \cos \psi \\ \dot{\psi} &= r \\ \dot{u} &= \frac{1}{m - X_{\dot{u}}} [X_u u + X_{u|u}|u| + \tau_X] \\ \dot{v} &= \frac{1}{m - Y_{\dot{v}}} [-(m - X_{\dot{u}})ur + Y_v v + Y_{v|v}|v|] \\ \dot{r} &= \frac{1}{I_{zz} - N_{\dot{r}}} [(X_{\dot{u}} - Y_{\dot{v}})uv + N_r r + N_{r|r}|r| + \tau_N]\end{aligned}\quad (2)$$

Here, x, y and ψ refer to the spatial configuration, while u, v and r denote the velocity components in the body-attached frame. $X_{\dot{u}}, Y_{\dot{v}}$, and $N_{\dot{r}}$ quantify added mass effects, while X_u, Y_v , and N_r correspond to linear damping coefficients. Quadratic damping terms $X_{u|u}, Y_{v|v}$, and $N_{r|r}$ capture nonlinear hydrodynamic resistance. The control inputs τ_X and τ_N govern surge propulsion and yaw torque respectively, with m representing total mass and I_{zz} the vertical axis moment of inertia. All hydrodynamic coefficients were obtained through CFD simulations, with the numerical values and detailed procedures provided in Reference [14].

C. Model Predictive Control

At each sampling instant t , the Model Predictive Control (MPC) solves the following finite-horizon optimal control problem over a prediction horizon of length N :

$$\begin{aligned}&\underset{\tau_{t|t}, \dots, \tau_{t+N-1|t}}{\text{minimize}} && \sum_{k=0}^{N-1} \ell(\chi_{t+k|t}, \tau_{t+k|t}), \\ &\text{subject to} && \chi_{t+k+1|t} = \chi_{t+k|t} + \Delta t \cdot f(\chi_{t+k|t}, \tau_{t+k|t}), \\ & && k = 0, \dots, N-1, \\ & && \chi_{t|t} = \chi(t), \\ & && \tau_{t+k|t} \in \chi, \quad k = 0, \dots, N-1, \\ & && \chi_{t+k|t} \in X, \quad k = 0, \dots, N-1,\end{aligned}\quad (3)$$

where $f(\chi, \tau)$ denotes continuous-time dynamics, $\tau \in \mathbb{R}^2$ and $\chi \in \mathbb{R}^6$ are control inputs and states respectively, and $\ell(\chi, \tau)$ is defined as:

$$\begin{aligned}\ell(\chi, \tau) &= Q_\psi (\psi - \psi_{ref})^2 + \tau^T R \tau \\ &\quad + (\tau - \tau_{prev})^T R_\Delta (\tau - \tau_{prev})\end{aligned}\quad (4)$$

with the following weighting matrices and parameters: $Q_\psi = 100$, $R = \text{diag}(1, 1)$ and $R_\Delta = \text{diag}(5, 1)$. After solving the optimization, only the first control input $\tau_{t|t}$ is applied to the system. The computational optimization process is accelerated by CasADi [27].

III. SYSTEM IDENTIFICATION

This section presents the system identification method based on Koopman theory and its online implementation via a Kalman filter.

A. The Koopman Operator

The Koopman operator advances measurement functions of the state with the flow of the dynamics. The Koopman operator $\mathcal{K}_t$ is an infinite dimensional linear operator that acts on eigenfunction ξ as:

$$\mathcal{K}_t \xi = \xi \circ F_t \quad (5)$$

where $\circ$ is the composition operator, t refers to the time step and F_t denotes a continuously mapping function.

With the Extended Dynamic Mode Decomposition (EDMD) method [9], the necessary observables are first selected to construct a finite dimensional space, followed by the extraction of eigenvalues from the data, thereby obtaining a finite dimensional approximate linear model. It should be noted that this model is not unique, and its specific form is entirely contingent upon the choice of observables.

Generally, the selection of observables primarily comprises three components: system states, inputs, and their associated functions. In multi-step prediction problems, to eschew introducing unnecessary complexity, the coupling functions between system states and inputs are typically not considered as independent observables. Based on this rationale, the evolutionary process of the system can be formulated as:

$$f[k+1] = \Theta^T \cdot f[k] \quad (6)$$

where $f[k]$ is the observables vector and Θ^T denotes the coefficient matrix that needs to be identified from data

The central goal of system identification is to minimize the fitting error, which can be mathematically formulated as minimizing the L^2 norm of the fitting error over all data.

$$\min_{\Theta} \sum_{k=1}^K \|\Theta^T \cdot f[k] - f[k+1]\|_2^2 \quad (7)$$

A linear koopman approximate model of the nonlinear system can be established by solving this least squares problem through methods such as singular value decomposition (SVD).

Path following is a common task for mobile robots, where the Line of Sight (LOS) method is widely applied to underactuated systems due to its ability to generate desired heading angles via waypoints, thereby reducing control complexity [4]. The heading degree of freedom is chosen as the control entry point, mainly based on the following two reasons: first, within the LOS framework, path following can be achieved solely through control of the heading degree of freedom; second, the key state variables corresponding to the heading degree of freedom—including the heading angle ψ and turning rate r —can be directly obtained from a single IMU sensor.

Based on the above considerations, this letter focuses on the heading degree of freedom for the control system design. To construct the system state observation framework, the state vector, control input, and reference signal at the current time step are defined as $\chi[k] = (\chi_x[k], \chi_y[k], \chi_\psi[k], \chi_u[k], \chi_v[k], \chi_r[k])^T \in \mathbb{R}^6$, $\tau_N[k] \in \mathbb{R}$, and $\zeta[k] \in \mathbb{R}$, respectively. Here, $\chi_x[k], \chi_y[k]$ denote the planar positions, $\chi_\psi[k]$ is the heading angle, and $\chi_u[k], \chi_v[k], \chi_r[k]$ represent the surge, sway, and yaw velocities. Subsequently, the lifted observable function vector $f[k]$ is constructed as follows:

$$f[k] = \begin{bmatrix} \chi_x[k], \chi_y[k], \chi_\psi[k], \chi_u[k], \chi_v[k], \chi_r[k], \\ \varepsilon[k], \chi_r[k]^2, \tau_N[k], \tau_N[k]^2 \end{bmatrix} \in \mathbb{R}^{10} \quad (8)$$

where $\varepsilon[k] = \zeta[k] - \chi_\psi[k]$ is the tracking error. Compared to generic basis functions such as radial basis functions (RBFs) or monomials, the proposed lifting functions are specifically tailored to the system's control objectives and excitation characteristics. This design enhances the Koopman model's predictive accuracy and interpretability without introducing unnecessary complexity or overfitting risks.

效果:

$$\begin{aligned}\dot{x} &= u \cos \psi - v \sin \psi \\ \dot{y} &= u \sin \psi + v \cos \psi \\ \dot{\psi} &= r \\ \dot{u} &= \frac{1}{m-X_{\dot{u}}} [X_u u + X_{u|u}|u| + \tau_X] \\ \dot{v} &= \frac{1}{m-Y_{\dot{v}}} [-(m-X_{\dot{u}})ur + Y_v v + Y_{v|v}|v|] \\ \dot{r} &= \frac{1}{I_{zz}-N_{\dot{r}}} [(X_{\dot{u}}-Y_{\dot{v}})uv + N_r r + N_{r|r}|r| + \tau_N]\end{aligned}\quad (2)$$

这里, x, y 和 ψ 指空间配置, 而 u, v 和 r 表示在机体附着坐标系中的速度分量。 $X_{\dot{u}}, Y_{\dot{v}}$ 和 $N_{\dot{r}}$ 用于量化附加质量效应, 而 X_u, Y_v 和 N_r 对应线性阻尼系数。二次阻尼项 $X_{u|u}, Y_{v|v}$ 和 $N_{r|r}$ 捕捉非线性水动力阻力。控制输入 τ_X 和 τ_N 分别控制纵向推进和偏航力矩, m 表示总质量, I_{zz} 为垂直轴转动惯量。所有水动力系数均通过 CFD 模拟获得, 数值和详细过程见参考文献 [14]。

C. Model Predictive Control

在每个采样时刻 t , 模型预测控制 (MPC) 解决以下在预测时域长度为 N 的有限时域最优控制问题:

$$\begin{aligned}\text{minimize} \quad & \sum_{k=0}^{N-1} \ell(\chi_{t+k|t}, \tau_{t+k|t}), \\ \text{subject to} \quad & \chi_{t+k+1|t} = \chi_{t+k|t} + \Delta t \cdot f(\chi_{t+k|t}, \tau_{t+k|t}), \\ & k = 0, \dots, N-1, \\ & \chi_{t|t} = \chi(t), \\ & \tau_{t+k|t} \in \mathcal{X}, \quad k = 0, \dots, N-1, \\ & \chi_{t+k|t} \in X, \quad k = 0, \dots, N-1,\end{aligned}\quad (3)$$

其中 $f(\chi, \tau)$ 表示连续时间动力学, $\tau \in \mathbb{R}^2$ 和 $\chi \in \mathbb{R}^6$ 分别是控制输入和状态, $\ell(\chi, \tau)$ 定义为:

$$\begin{aligned}\ell(\chi, \tau) &= Q_\psi (\psi - \psi_{ref})^2 + \tau^T R \tau \\ &+ (\tau - \tau_{prev})^T R_\Delta (\tau - \tau_{prev})\end{aligned}\quad (4)$$

使用以下权重矩阵和参数: $Q_\psi = 100$, $R = \text{diag}(1, 1)$ 和 $R_\Delta = \text{diag}(5, 1)$ 。在求解优化问题后, 只有第一个控制输入 $\tau_{t|t}$ 被应用到系统中。计算优化过程由 CasADi [27] 加速。

三、系统识别

本节介绍了基于库普曼理论的系统辨识方法及其通过卡尔曼滤波器的在线实现。

A. The Koopman Operator

库普曼算子随着动力学的流推动状态的测量函数。库普曼算子 $\mathcal{K}_t$ 是一个作用于本征函数 ξ 的无限维线性算子, 其作用为:

$$\mathcal{K}_t \xi = \xi \circ F_t \quad (5)$$

其中 $\circ$ 是组合运算符, t 指时间步长, F_t 表示一个连续映射函数。

使用扩展动态模态分解 (EDMD) 方法[9], 首先选择必要的观测量来构建一个有限维空间, 然后从数据中提取特征值, 从而获得一个有限维的近似线性模型。需要注意的是, 该模型并非唯一, 其具体形式完全取决于观测量的选择。

通常, 可观测量的选择主要包括三个组成部分: 系统状态、输入及其相关函数。在多步预测问题中, 为了避免引入不必要的复杂性, 系统状态与输入之间的耦合函数通常不被视为独立的可观测测量。基于这一原理, 系统的演化过程可以表述为:

$$f[k+1] = \Theta^T \cdot f[k] \quad (6)$$

其中 $f[k]$ 是观测向量, Θ^T 表示需要从数据中确定的系数矩阵

系统识别的核心目标是最小化拟合误差, 这可以用数学形式表示为在所有数据上最小化拟合误差的 L^2 范数。

$$\min_{\Theta} \sum_{k=1}^K \|\Theta^T \cdot f[k] - f[k+1]\|_2^2 \quad (7)$$

可以通过奇异值分解 (SVD) 等方法求解此最小二乘问题, 从而建立非线性系统的线性Koopman近似模型。

路径跟随是移动机器人一个常见的任务, 其中视线 (LOS) 方法广泛应用于欠驱动系统, 因为它能够通过路径点生成期望航向角, 从而降低控制复杂性 [4]。航向自由度被选择作为控制切入点, 主要基于以下两个原因: 首先, 在LOS框架下, 仅通过控制航向自由度即可实现路径跟随; 其次, 与航向自由度对应的关键状态变量——包括航向角 ψ 和转向速率 r ——可以直接从单个IMU传感器获取。

基于以上考虑, 本信主要关注控制系统设计中的航向自由度。为了构建系统状态观测框架, 当前时间步的状态向量、控制输入和参考信号分别定义为 $\chi[k] = (\chi_x[k], \chi_y[k], \chi_\psi[k], \chi_u[k], \chi_v[k], \chi_r[k])^T \in \mathbb{R}^6$, $\tau_N[k] \in \mathbb{R}$, 以及 $\zeta[k] \in \mathbb{R}$ 。这里, $\chi_x[k], \chi_y[k]$ 表示平面位置, $\chi_\psi[k]$ 为航向角, $\chi_u[k], \chi_v[k], \chi_r[k]$ 表示前进、横向和偏航速度。随后, 提升的可观测函数向量 $f[k]$ 构建如下:

$$f[k] = \begin{bmatrix} \chi_x[k], \chi_y[k], \chi_\psi[k], \chi_u[k], \chi_v[k], \chi_r[k], \\ \varepsilon[k], \chi_r[k]^2, \tau_N[k], \tau_N[k]^2 \end{bmatrix} \in \mathbb{R}^{10} \quad (8)$$

其中 $\varepsilon[k] = \zeta[k] - \chi_\psi[k]$ 是跟踪误差。与径向基函数 (RBFs) 或单项式等通用基函数相比, 所提出的提升函数是专门针对系统的控制目标和激励特性设计的。这种设计提高了Koopman模型的预测准确性和可解释性, 同时不会引入不必要的复杂性或过拟合风险。

B. Online Identification Algorithm Based on the Kalman Filter

The coefficient matrix Θ in (6) varies over time in time varying systems. Conventional state estimation methods typically assume constant model parameters and employ the Kalman filter for state estimation. Given the assumption that the system state $\chi[k+1]$ acquired by sensors is sufficiently accurate, the Kalman filter can achieve optimal estimation of the parameter matrix Θ under noisy conditions based on observation data, namely the function $f[k+1]$ constructed from $\chi[k+1]$, thereby accomplishing online parameter identification. The design objective of the Kalman filter is to minimize the square of the posterior estimation error, which aligns with the optimization goal of online identification. Following the approach in reference [18], a separate Kalman filter is constructed for each equation to enable parallel online identification. The prediction model and observation function are as follows:

$$\text{Prediction: } \vartheta_i[k+1] = \vartheta_i[k] \quad (9)$$

$$\text{Observation: } f_i[k+1] = \vartheta_i^T[k+1]\mathbf{f}[k] \quad (10)$$

where ϑ_i is coefficient vector in Θ , $i = 1, \dots, N$, and N is the variable about system states. Generally speaking, Θ changes slowly and can be considered to be constant in a timestep. Therefore, the online system identification based on the Kalman filter can be finished as:

$$\vartheta_i[k+1|k] = \vartheta_i[k|k] \quad (9a)$$

$$P_i[k+1|k] = P_i[k|k] \quad (9b)$$

$$K_i[k+1] = \frac{P_i[k+1|k]\mathbf{f}[k+1]}{\mathbf{f}^T[k+1]P_i[k+1|k]\mathbf{f}[k+1] + R_i[k]} \quad (9c)$$

$$e_i[k+1] = f_i[k+1] - \vartheta_i^T[k+1|k]\mathbf{f}[k] \quad (9d)$$

$$\vartheta_i[k+1|k+1] = \vartheta_i[k+1|k] + K_i[k+1]e_i[k+1] \quad (9e)$$

$$P_i[k+1|k+1] = (I - K_i[k+1]\mathbf{f}^T[k+1])P_i[k|k] \quad (9f)$$

where P_i is the covariance matrix, K_i denotes the Kalman gain vector, R_i refers the variance, and e_i indicates the prediction error. To cope with the time-varying nature of noise statistics in real-world environments, an adaptive strategy is employed. The process noise covariance Q and measurement noise covariance R are initially tuned using offline data; during online operation, R is recursively estimated based on the innovation covariance, while Q is dynamically adjusted according to the trend of prediction errors, thereby maintaining optimal estimation performance. The complete update procedure of the Koopman model parameter Θ is detailed in the pseudocode provided in Algorithm 1.

IV. OFFLINE KOOPMAN MODEL IDENTIFICATION

This section outlines the data driven identification and subsequent validation of a dynamic model for a bio-inspired turtle robot, leveraging Koopman operator theory.

A. Data Collection and System Identification

Data driven methods for dynamic model identification place high demands on data quality. To meet this requirement, this study utilizes the dynamic simulator Webots to acquire high fidelity motion data of a bio-inspired turtle robot. To generate a

Algorithm 1: Online Koopman Model Identification via Kalman Filter.

Require: Offline identified parameters $\vartheta_i[0]$, experimental data from MPC tracking, forgetting factor λ , scaling factor α , adjustment period L , bounds $[Q_{\min}, Q_{\max}]$

Ensure: Updated Koopman model parameter matrix Θ

- 1: Initialize $P_i[0] \leftarrow I$, $Q \leftarrow Q_0$, $R \leftarrow R_0$, $\mathbf{f}[0]$, $S_{\text{innov}}[0] \leftarrow \mathbf{0}$
- 2: $K_T \leftarrow$ total time steps, $N \leftarrow 10$
- 3: **for** $k = 0$ to $K_T - 1$ **do**
- 4: Read new measurements $\chi[k+1]$, $\zeta[k+1]$, $\tau[k+1]$
- 5: Compute $\varepsilon[k+1] \leftarrow \zeta[k+1] - \chi_\psi[k+1]$
- 6: Construct $\mathbf{f}[k+1] \leftarrow (8)$
- 7: Innovation: $\nu[k+1] \leftarrow \mathbf{y}[k+1] - \hat{\mathbf{y}}$
- 8: Update innovation covariance: $S_{\text{innov}}[k+1] \leftarrow \lambda S_{\text{innov}}[k] + (1 - \lambda)\nu[k+1]\nu[k+1]^T$
- 9: Adapt measurement noise:
 $R \leftarrow \alpha \text{diag}(S_{\text{innov}}[k+1])$ (if $k < 10$, keep $R = R_0$)
- 10: **for** $i = 1$ to N **do**
- 11: Kalman filter update for ϑ_i using 9
- 12: Update $\Theta[:, i]$
- 13: **end for**
- 14: Periodic Q adaptation: if $\text{mod}(k+1, L) = 0$, compute recent error trend Δe ; if $\Delta e > \delta$, $Q \leftarrow \min(\beta Q, Q_{\max})$; if $\Delta e < -\delta$, $Q \leftarrow \max(\gamma Q, Q_{\min})$.
- 15: $\mathbf{f}[k] \leftarrow \mathbf{f}[k+1]$
- 16: **for** $i = 1$ to N **do**
- 17: $\vartheta_i[k|k] \leftarrow \vartheta_i[k+1|k+1]$
- 18: $P_i[k|k] \leftarrow P_i[k+1|k+1]$
- 19: **end for**
- 20: **end for**

dataset that sufficiently excites the system's dynamic characteristics, path following task and heading angle tracking task are designed. The specific tasks are illustrated in Fig. 2(a).

Using this dataset, an offline Koopman model is identified by solving the least squares problem described in (7) with the lifting functions defined in (8). To validate the open-loop prediction accuracy of the model, the representative horizontally aligned figure "8" path is selected as the validation set. As a benchmark for comparison, the parameters of the dynamic model given in (2) in Section III are also directly identified using the least squares method based on the same dataset, following the strategy consistent with Reference [28]. Finally, a comparative analysis of the open loop heading angle predictions from the parametric CFD model, the least squares identified model, and the Koopman model is presented in Fig. 3. The reference signal serves as an external input in the Koopman model, whereas the baseline methods are driven solely by the initial state and control inputs.

Furthermore, to comprehensively evaluate the generalization capability of the model, predictions were made for the heading angle variations across four prediction sets, including the horizontally aligned figure "8" path, and the results are presented in Fig. 3(c). The comparison demonstrates that the proposed Koopman model achieves significantly higher prediction accuracy than the other two baseline models. For instance, in predicting the heading angle for a heartshaped trajectory, the average

B. Online Identification Algorithm Based on the Kalman Filter

在时间可变系统中, (6)中的系数矩阵 Θ 随时间变化。传统状态估计方法通常假设模型参数为常数, 并采用卡尔曼滤波器进行状态估计。假设传感器获取的系统状态 $\chi[k+1]$ 足够精确, 卡尔曼滤波器能够在噪声条件下基于观测数据对参数矩阵 Θ 实现最优估计, 即从 $\chi[k+1]$ 构建的函数 $f[k+1]$, 从而完成在线参数辨识。卡尔曼滤波器的设计目标是最小化后验估计误差的平方, 这与在线辨识的优化目标一致。按照参考文献 [18] 的方法, 为每个方程构建一个独立的卡尔曼滤波器, 以实现并行在线辨识。预测模型和观测函数如下:

$$\text{Prediction: } \vartheta_i[k+1] = \vartheta_i[k] \quad (9)$$

$$\text{Observation: } f_i[k+1] = \vartheta_i^T[k+1]\mathbf{f}[k] \quad (10)$$

其中 ϑ_i 是 Θ 中的系数向量, $i = 1, \dots, N$, N 是关于系统状态的变量。一般来说, Θ 变化缓慢, 并且可以在一个时间步内认为是常数。因此, 基于卡尔曼滤波器的在线系统辨识可以完成如下:

$$\vartheta_i[k+1|k] = \vartheta_i[k|k] \quad (9a)$$

$$P_i[k+1|k] = P_i[k|k] \quad (9b)$$

$$K_i[k+1] = \frac{P_i[k+1|k]\mathbf{f}[k+1]}{\mathbf{f}^T[k+1]P_i[k+1|k]\mathbf{f}[k+1] + R_i[k]} \quad (9c)$$

$$e_i[k+1] = f_i[k+1] - \vartheta_i^T[k+1|k]\mathbf{f}[k] \quad (9d)$$

$$\vartheta_i[k+1|k+1] = \vartheta_i[k+1|k] + K_i[k+1]e_i[k+1] \quad (9e)$$

$$P_i[k+1|k+1] = (I - K_i[k+1]\mathbf{f}^T[k+1])P_i[k|k] \quad (9f)$$

其中 P_i 是协方差矩阵, K_i 表示卡尔曼增益向量, R_i 指方差, e_i 表示预测误差。为了应对实际环境中噪声统计量的时变特性, 采用了一种自适应策略。过程噪声协方差 Q 和测量噪声协方差 R 初始通过离线数据进行调节; 在在线运行过程中, R 基于创新协方差递归估计, 而 Q 则根据预测误差的趋势动态调整, 从而保持最优估计性能。Koopman 模型参数 Θ 的完整更新过程在算法 1 中提供的伪代码中详细说明。

IV. 离线库普曼模型识别

本节概述了利用库普曼算子理论, 对仿生海龟机器人进行数据驱动的动态模型识别及随后验证的过程。

A. Data Collection and System Identification

用于动态模型识别的数据驱动方法对数据质量要求很高。为了满足这一要求, 本研究利用动态模拟器 Webots 获取仿生海龟机器人的高精度运动数据。为了生成一个

算法 1: 通过在线 Koopman 模型识别

Kalman Filter.

Require: Offline identified parameters $\vartheta_i[0]$, experimental data from MPC tracking, forgetting factor λ , scaling factor α , adjustment period L , bounds $[Q_{\min}, Q_{\max}]$

Ensure: Updated Koopman model parameter matrix Θ

```

1: Initialize  $P_i[0] \leftarrow I, Q \leftarrow Q_0, R \leftarrow R_0, \mathbf{f}[0], S_{\text{innov}}[0] \leftarrow \mathbf{0}$ 
2:  $K_T \leftarrow$  total time steps,  $N \leftarrow 10$ 
3: for  $k = 0$  to  $K_T - 1$  do
4:   Read new measurements  $\chi[k+1], \zeta[k+1], \tau[k+1]$ 
5:   Compute  $\varepsilon[k+1] \leftarrow \zeta[k+1] - \chi_\psi[k+1]$ 
6:   Construct  $\mathbf{f}[k+1] \leftarrow (8)$ 
7:   Innovation:  $\nu[k+1] \leftarrow \mathbf{y}[k+1] - \hat{\mathbf{y}}$ 
8:   Update innovation covariance:  $S_{\text{innov}}[k+1] \leftarrow \lambda S_{\text{innov}}[k] + (1 - \lambda)\nu[k+1]\nu[k+1]^T$ 
9:   Adapt measurement noise:
      $R \leftarrow \alpha \text{diag}(S_{\text{innov}}[k+1])$  (if  $k < 10$ , keep  $R = R_0$ )
10:  for  $i = 1$  to  $N$  do
11:    Kalman filter update for  $\vartheta_i$  using 9
12:    Update  $\Theta[:, i]$ 
13:  end for
14:  Periodic  $Q$  adaptation: if  $\text{mod}(k+1, L) = 0$ ,
    compute recent error trend  $\Delta e$ ; if  $\Delta e > \delta$ ,
     $Q \leftarrow \min(\beta Q, Q_{\max})$ ; if  $\Delta e < -\delta$ ,
     $Q \leftarrow \max(\gamma Q, Q_{\min})$ .
15:   $\mathbf{f}[k] \leftarrow \mathbf{f}[k+1]$ 
16:  for  $i = 1$  to  $N$  do
17:     $\vartheta_i[k|k] \leftarrow \vartheta_i[k+1|k+1]$ 
18:     $P_i[k|k] \leftarrow P_i[k+1|k+1]$ 
19:  end for
20: end for

```

为了充分激发系统的动态特性, 设计了路径跟随任务和航向角跟踪任务。具体任务如图2(a)所示。

使用该数据集, 通过求解 (7) 中描述的最小二乘问题并使用 (8) 中定义的提升函数, 识别出一个离线 Koopman 模型。为了验证模型的开环预测精度, 选择具有代表性的水平8字形路径作为验证集。作为对比基准, 第III节 (2) 中给出的动态模型参数也使用相同数据集并遵循与参考文献 [28] 一致的策略, 通过最小二乘方法直接识别。最后, 在图3中展示了参数化 CFD 模型、最小二乘识别模型和 Koopman 模型的开环航向角预测的比较分析。在 Koopman 模型中, 参考信号作为外部输入, 而基线方法仅由初始状态和控制输入驱动。

此外, 为了全面评估模型的泛化能力, 对四个预测集的航向角变化进行了预测, 包括水平排列的“8”字形路径, 结果如图3(c)所示。比较表明, 所提出的 Koopman 模型的预测精度显著高于另外两个基线模型。例如, 在预测心形轨迹的航向角时, 平均值

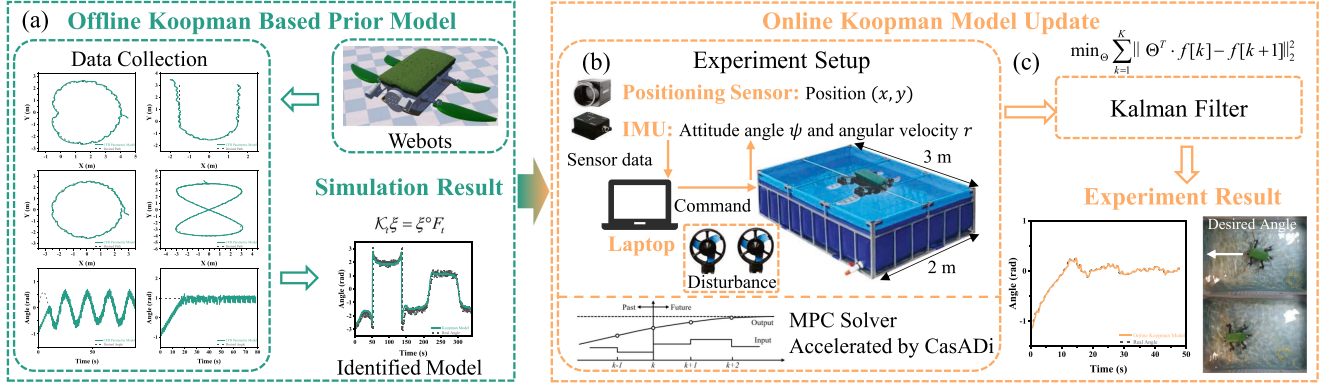


Fig. 2. Koopman operator based Sim2Real transfer framework. (a) Offline Koopman model identification from simulation data. (b) Experimental setup for real-world data acquisition. Location data can be acquired through various sensors, such as GPS for surface applications or acoustic equipment for underwater positioning. In this letter, a camera was utilized in convenience of experimental setup. (c) Online Koopman model updating via experimental data and a Kalman filter.

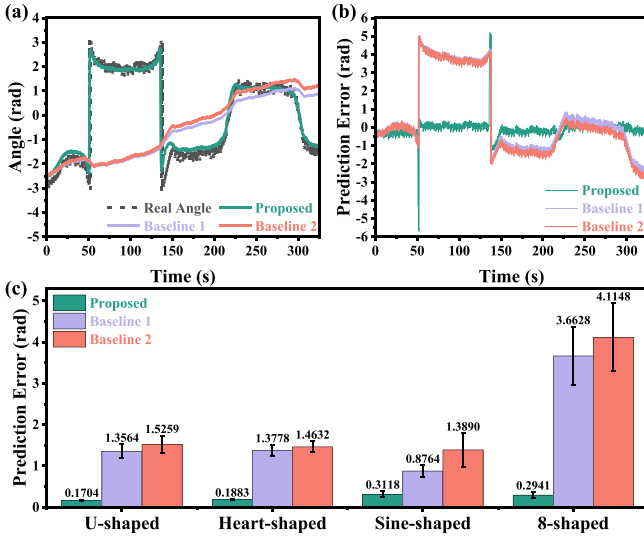


Fig. 3. Analysis of the offline model accuracy. (a) Comparison of heading angle predictions among different identification models in figure-eight path following. (b) Comparison of heading angle prediction errors for the corresponding models. (c) Comparison of heading angle prediction errors for different data type, with error bars calculated from five simulation trials. Proposed: Koopman Model. Baseline 1: CFD Parametric Model. Baseline 2: Least Square Identified Model.

prediction error of the Koopman model is only 0.1883 rad, which is substantially lower than the 1.3778 rad of the CFD based parametric model and the 1.4632 rad of the least squares identified model.

The results indicate that only the Koopman model can accurately identify the abrupt changes in the robot's yaw angle during motion and satisfactorily track its variation trend. In contrast, the predictions from the parametric CFD based model and the least squares identified model exhibit smoother variations, suggesting their difficulty in capturing the dynamic transients of the system's motion.

B. Validation of the Generalization Capability

To effectively validate that the proposed method encodes the essential dynamic behavior of the turtle-inspired robot and

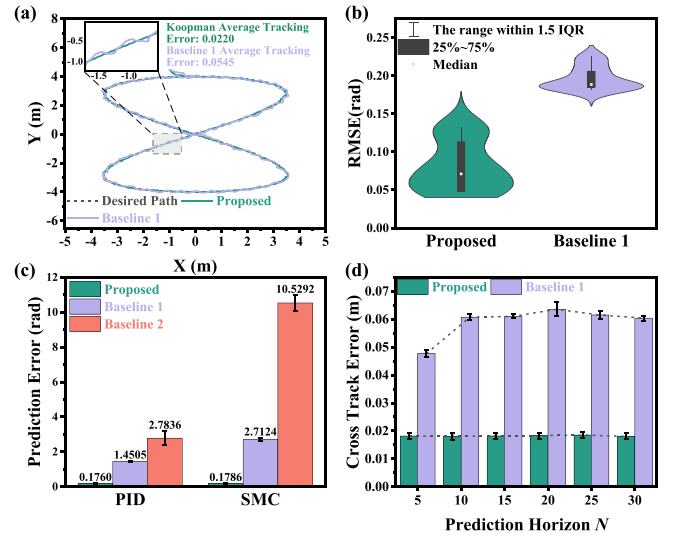


Fig. 4. Validation of the generalization capability. (a) Trajectory comparison between the proposed method and baseline method in figure-eight path following. (b) Violin plots of MPC tracking error comparison between the proposed method and baseline method under random reference signals. (c) Prediction comparison of the proposed method and baseline method for trajectories generated by different controllers, with error bars calculated from five simulation trials. (d) Tracking error comparison between the proposed method and baseline method under different prediction horizons, with error bars calculated from five simulation trials. Proposed: Koopman Model. Baseline 1: CFD Parametric Model. Baseline 2: Least Square Identified Model.

generalizes well across varying reference inputs and control laws, simulation analysis is conducted as shown in Fig. 4. Under an identical control framework and parameter set, the tracking trajectory of the proposed method is smoother; quantitative analysis reveals an average lateral tracking error of 0.0220 m, which is significantly lower than the 0.0545 m error of the parametric CFD model, achieving an improvement of accuracy by 59.6%.

Following the principle of Pseudo-Random Binary Sequence (PRBS), we generated diverse reference signals and used the same MPC controller to track them. The results demonstrate a narrower interquartile range and a more concentrated error distribution, indicating that Koopman model captures intrinsic

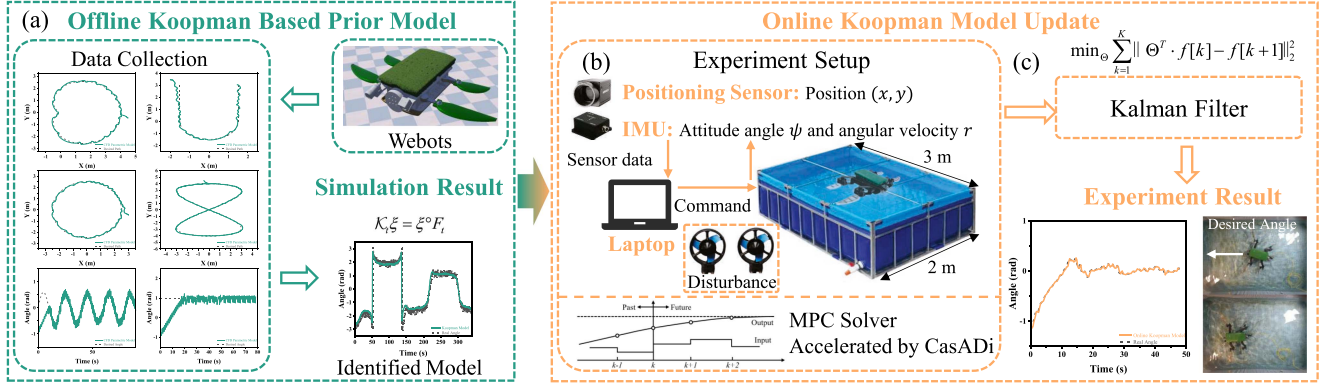


图2. 基于 Koopman 算子的 Sim2Real 转移框架。(a) 从仿真数据进行的离线 Koopman 模型识别。(b) 实验装置用于真实世界数据采集。位置数据可以通过各种传感器获取，例如用于地表应用的 GPS 或用于水下定位的声学设备。在本信中，为了实验装置的便利，使用了相机。(c) 通过实验数据和卡尔曼滤波器进行 Koopman 模型的在线更新。

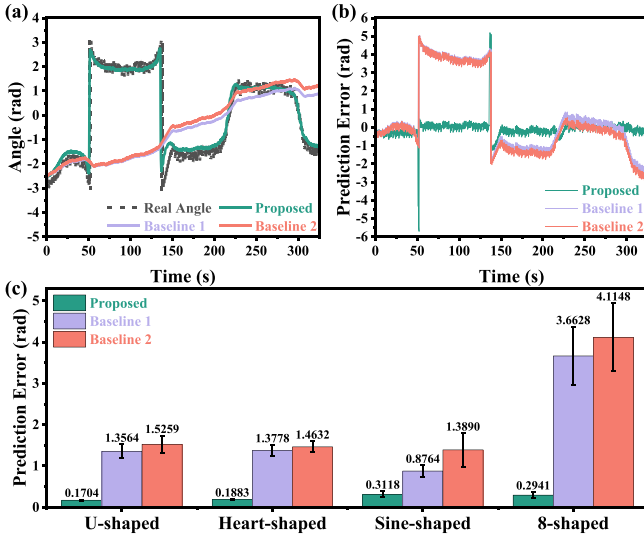


图3. 离线模型精度分析。(a) 图八路径跟踪中不同识别模型的航向角预测比较。(b) 相应模型的航向角预测误差比较。(c) 不同数据类型航向角预测误差比较，误差棒由五次仿真试验计算得出。提出的模型：Koopman 模型。基线 1：CFD 参数模型。基线 2：最小二乘识别模型。

Koopman 模型的预测误差仅为 0.1883 弧度，这显著低于基于 CFD 的参数模型的 1.3778 弧度和最小二乘识别模型的 1.4632 弧度。

结果表明，只有 Koopman 模型能够准确识别机器人运动过程中偏航角的突变，并令人满意地跟踪其变化趋势。相比之下，基于参数 CFD 的模型和最小二乘识别模型的预测变化更加平滑，这表明它们在捕捉系统运动的动态瞬态方面存在困难。

B. Validation of the Generalization Capability

为了有效验证所提出的方法是否编码了受乌龟启发的机器人的基本动态行为，并且

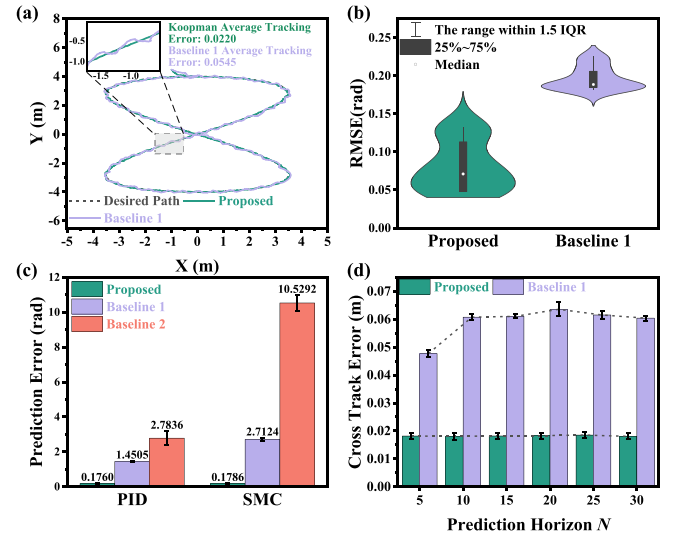


图4. 泛化能力验证。(a) 所提方法与基线方法在数字八字路径跟踪中的轨迹对比。(b) 在随机参考信号下，所提方法与基线方法的 MPC 跟踪误差小提琴图对比。(c) 所提方法与基线方法对不同控制器生成轨迹的预测对比，误差棒由五次仿真试验计算得出。(d) 在不同预测时间范围内，所提方法与基线方法的跟踪误差对比，误差棒由五次仿真试验计算得出。所提方法：Koopman 模型。基线 1：CFD 参数化。基线 2：最小二乘识别模型。

模型。基线 2：最小二乘识别模型。

为了在不同的参考输入和控制律下实现良好泛化，如图4所示进行了仿真分析。在相同的控制框架和参数设置下，所提出方法的跟踪轨迹更加平滑；定量分析显示，其平均横向跟踪误差为 0.0220 m，显著低于参数化 CFD 模型的 0.0545 m 误差，精度提高了 59.6%。

遵循伪随机二进制序列 (PRBS) 的原理，我们生成了多样的参考信号，并使用相同的 MPC 控制器对其进行跟踪。结果显示四分位间距更窄，误差分布更集中，这表明 Koopman 模型捕捉到了内在特性

dynamic features rather than memorizing reference-specific behaviors, as shown in the violin plot of Fig. 4(b). To further assess generalization across different controllers, we employed the learned Koopman model to predict robot motion under two unseen control strategies: PID and Sliding Mode Control (SMC). As illustrated in Fig. 4(c), the proposed method maintained superior prediction performance in both cases, confirming that the model is not overfitted to a particular controller, but instead captures the underlying open-loop dynamics of the robotic system.

To analyze the impact of the model's long-horizon prediction accuracy on the controller, this letter examines the path-tracking precision of the proposed method and the baseline method under different prediction horizons, with results presented in Fig. 4(d). The average cross tracking error of the proposed method remains consistently stable at approximately 0.018 m, indicating that the error does not significantly accumulate with the extension of the prediction horizon. In contrast, for the baseline method, the tracking error rises markedly as N increases from 5 to 10, directly reflecting that model mismatch exacerbates the error as the prediction horizon expands, thereby reducing the control performance of the MPC.

V. ONLINE KOOPMAN MODEL UPDATE

This section details the experimental implementation of an online Kalman filter (KF) based update mechanism for the Koopman model.

A. Experiment Setup and Online Model Update

Simulation data often fails to capture the full complexity of the real world, particularly the influence of environmental disturbances such as waves and flows on robot dynamics. To address this issue and achieve rapid acquisition of a high precision model, this study introduces an online update mechanism built upon a prior model established from simulation data.

The experimental setup is shown in Fig. 2(b). The robot's pose is obtained through multi sensor fusion: an overhead camera fixed above the water pool provides positional coordinates, from which the velocity is derived via differentiation; an onboard IMU directly measures the yaw angle and yaw rate. To evaluate the real time predictive capability of the online updated model, we employ a one step ahead prediction to test the model performance. The primary objective is to determine whether the final model after online updating outperforms the original offline model. The Recursive Least Squares (RLS) algorithm is selected as a benchmark for comparison. This method achieves parameter updates recursively through a mathematical transformation. In contrast, the Kalman filter based update method adopted in this work essentially minimizes the posterior estimation error.

As outlined in Algorithm 1 and Fig. 2(c), the online parameter update is performed. The results are presented in Fig. 5(a). The model updated via the Kalman filter responds rapidly and closely tracks the system's dynamic changes. In comparison, the original offline model and the model updated using RLS exhibit sluggish tracking and inadequate response. As shown in the logarithmic scale tracking error plot in Fig. 5(b), the proposed Kalman filter based update method achieves a significantly smaller mean prediction error of 0.0328 *rad*, compared to 0.0566 *rad* for the RLS method and 0.0988 *rad* for the offline model. In this study, five sets of directional tracking data were collected. The average

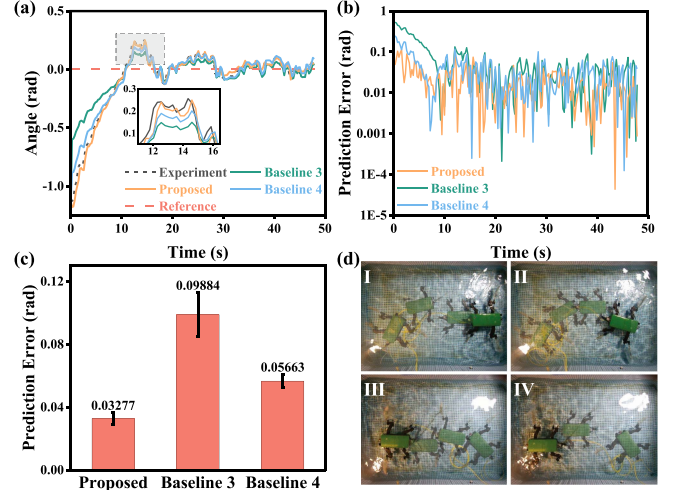


Fig. 5. Analysis of the online model accuracy. (a) Multi model comparison of one step predictions for heading tracking. (b) Comparison of yaw tracking error in logarithmic scale. (c) Online average prediction errors for various models on five experimental datasets, with error bars indicating the standard error. (d) Selected experimental snapshots. Proposed: Koopman Model + KF. Baseline 3: Offline Koopman Model. Baseline 4: Koopman Model + RLS.

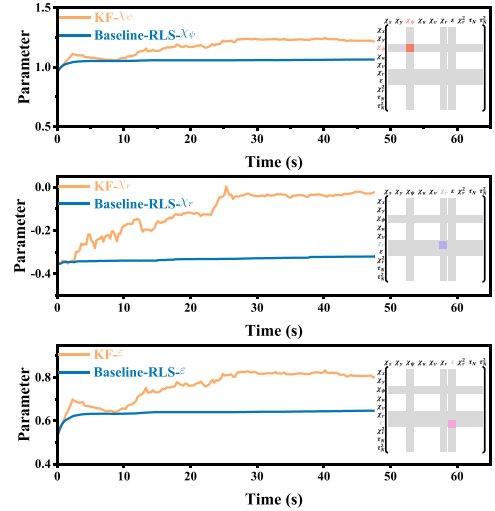


Fig. 6. Comparison of online updates for Key Koopman model parameters.

prediction errors are presented in Fig. 5(c). Compared with the RLS method, the proposed framework improves the prediction accuracy by 42.13%. Selected experimental snapshots are shown in Fig. 5(d).

Furthermore, Fig. 6 illustrates the temporal variation of key model parameters (ψ and r) during the yaw angle tracking task. The figure shows that the Kalman filter method applies more substantial updates to the parameters, and the parameter values converge to stability more quickly. This further validates the effectiveness of the Kalman filter method in rapidly converging to and stably estimating model parameters in dynamic environments.

B. Verification With MPC in Experiment

To evaluate model performance under disturbed conditions, this study conducted experiments in a water tank by arranging

动态特征,而不是记忆特定参考的行为,如图4(b)的小提琴图所示。为了进一步评估在不同控制器下的泛化能力,我们使用学习到的Koopman模型来预测机器人在两种未见控制策略下的运动:PID控制和滑模控制(SMC)。如图4(c)所示,该方法在这两种情况下都保持了较高的预测性能,验证了该模型并没有对特定控制器过拟合,而是捕捉到了机器人系统的潜在开环动力学。

为了分析模型长期预测精度对控制器的影响,本文考察了在不同预测时域下,所提出方法与基线方法的路径跟踪精度,结果如图4(d)所示。所提出方法的平均横向跟踪误差始终稳定在约0.018米,表明误差随预测时域的延长并未显著累积。相比之下,对于基线方法,当 N 从5增加到10时,跟踪误差显著上升,直接反映了随着预测时域扩大,模型不匹配加剧了误差,从而降低了MPC的控制性能。

V. 在线 Koopman 模型更新

本节详细说明了基于卡尔曼滤波器(KF)的Koopman模型在线更新机制的实验实现。

A. Experiment Setup and Online Model Update

模拟数据往往无法完全捕捉现实世界的复杂性,尤其是环境干扰(如波浪和流动)对机器人动力学的影响。为了解决这一问题并实现高精度模型的快速获取,本研究引入了一种基于从模拟数据建立的先验模型的在线更新机制。

实验装置如图2(b)所示。机器人的姿态是通过多传感器融合获得的:固定在水池上方的俯视相机提供位置坐标,通过微分从中推导速度;机载惯性测量单元(IMU)直接测量偏航角和偏航角速率。为了评估在线更新模型的实时预测能力,我们采用一步预测来测试模型性能。主要目标是确定在线更新后的最终模型是否优于原始离线模型。选择递归最小二乘(RLS)算法作为对照基准。该方法通过数学变换递归地实现参数更新。相比之下,本研究采用的基于卡尔曼滤波器的更新方法本质上是最小化后验估计误差。

如算法1和图2(c)所示,执行了在线参数更新。结果如图5(a)所示。通过卡尔曼滤波器更新的模型响应迅速,并能紧密追踪系统的动态变化。相比之下,原始的离线模型和使用RLS更新的模型表现出跟踪迟缓和响应不足。如图5(b)的对数刻度跟踪误差图所示,所提出的基于卡尔曼滤波器的更新方法实现了显著较小的平均预测误差,为0.0328 rad,而RLS方法为0.0566 rad,离线模型为0.0988 rad。在本研究中,收集了五组方向跟踪数据。平均

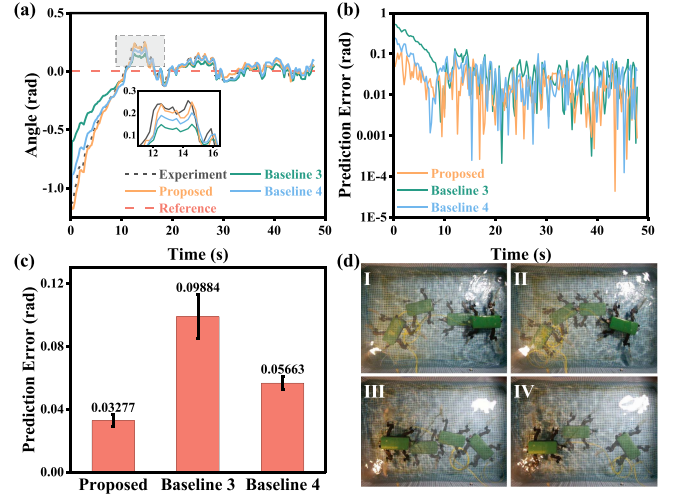


图5. 在线模型精度分析。(a) 多模型对航向跟踪一步预测的比较。(b) 偏航跟踪误差的对数尺度比较。(c) 五个实验数据集上各种模型的在线平均预测误差,误差线表示标准误差。(d) 选定的实验快照。提出模型:Koopman模型+KF。基线3:离线Koopman模型。基线4:Koopman模型+RLS。

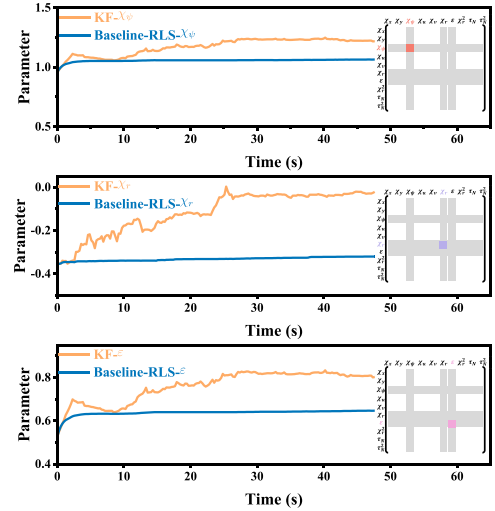


图6. Key Koopman模型参数的在线更新比较

S.

预测误差如图5(c)所示。与RLS方法相比,所提出的框架将预测精度提高了42.13%。选定的实验快照如图5(d)所示。

此外,图6展示了在偏航角跟踪任务中关键模型参数(ψ 和 τ)的时间变化。图中显示,卡尔曼滤波方法对参数进行了更为显著的更新,并且参数值更快地收敛到稳定状态。这进一步验证了卡尔曼滤波方法在动态环境中能够快速收敛并稳定估计模型参数的有效性。

B. Verification With MPC in Experiment

为了评估模型在扰动条件下的性能,本研究通过在水箱中进行实验并进行布置

TABLE II
ABLATION STUDY OF THE PROPOSED FRAMEWORK

Architecture	MAE (rad)	RMSE (rad)	MaxAE (rad)	STD (rad)	90% Percentile Error (rad)
Full Proposed	0.0328	0.0438	0.1346	0.0292	0.0723
Least Square + Sim2Real + KF	0.0581	0.0675	0.1666	0.0344	0.0992
Koopman + RLS	0.0566	0.0756	0.2487	0.0502	0.1229
Offline Koopman	0.0988	0.1402	0.4534	0.0986	0.2408

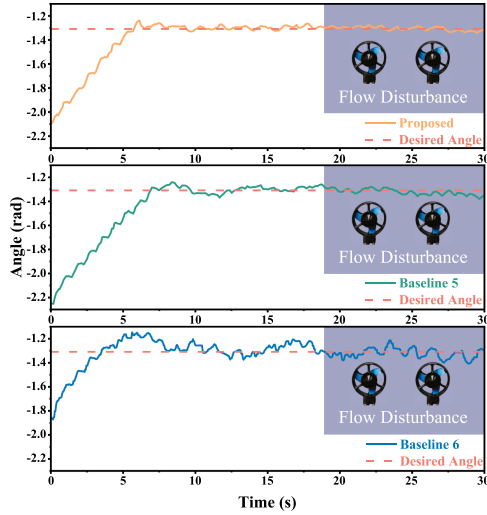


Fig. 7. Variations in the heading angle of the turtle inspired robot when performing directional tracking tasks. Proposed: MPC with online updated koopman model; Baseline 5: MPC with offline koopman model; Baseline 6: MPC with CFD parametric model.

propellers to generate controllable flow disturbances, thereby simulating the uncertain dynamics encountered in real-world operating conditions. To ensure comparability among different dynamic models, comparative experiments on orientation tracking based on MPC were carried out under consistent controller parameter settings.

The experimental results are shown in Fig. 7. Compared with the CFD parametric model, both the offline Koopman model and the online Koopman model exhibited lower overshoot and demonstrated better orientation maintenance performance under flow disturbances. Among them, the proposed online Koopman model further improved tracking stability, indicating its effectiveness in capturing unmodeled dynamics not represented in simulations. During the steady state tracking phase, the proposed method achieved a root mean square error (RMSE) of 0.0164 *rad*, which was lower than the 0.0294 *rad* of the offline Koopman model and the 0.0448 *rad* of the CFD parametric model. Compared with the baseline methods, the tracking error of the proposed approach was reduced by 63.4%, validating the effectiveness of the online update mechanism in enhancing model prediction accuracy and disturbance rejection capability. Fig. 8 displays a sequence of the robot's motion postures during the experiment, visually illustrating its movement and adjustments within the confined space.

C. Ablation Study

The ablation study results presented in Table II quantitatively demonstrate the contribution of each component within the proposed framework. The data in Table II are averaged over five

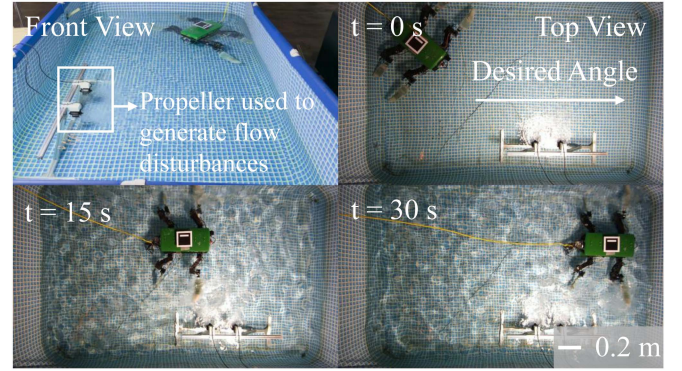


Fig. 8. Experimental snapshots of the turtle inspired robot during a directional tracking task.

experimental runs, with the selected metrics being mean absolute error (MAE), RMSE, maximum absolute error (MaxAE), standard deviation of error (STD), and the 90% percentile error. First, the necessity of online adaptation is evidenced by the fact that the Koopman model updated with RLS significantly outperforms the offline Koopman model, confirming that simulation data alone cannot fully capture real-world dynamics. Second, the superiority of the Kalman filter as an online update mechanism is validated by the full proposed method, which achieves the lowest errors across all metrics (MAE = 0.0328, RMSE = 0.0438), surpassing the Koopman+RLS variant. Finally, the importance of Koopman lifting is highlighted by comparing the full proposed method with the least squares-identified model: despite both employing Sim2Real pre-training and Kalman filtering, the Koopman-based architecture yields substantially lower prediction errors (e.g., MAE 0.0328 vs. 0.0581). These comparisons collectively affirm that each module—Koopman lifting, Sim2Real pre-training, and Kalman-based online update—indispensably contributes to the overall performance of the framework.

VI. CONCLUSION

In this letter, we propose an online system identification method based on the Koopman operator and integrate it within a Sim2Real transfer framework, which is successfully applied to the hydrodynamic modeling of a turtle inspired robotic system. The online algorithm, designed based on Kalman filter, enables rapid and accurate online updates of the offline trained prior model using limited real world experimental data, effectively capturing the time varying characteristics of the system in dynamic environments. Experimental results demonstrate that the proposed Koopman model outperforms traditional parametric CFD model and least squares identified methods in both yaw motion prediction and control tasks, while maintaining strong

表 II 提出的框架消融研究

Architecture	MAE (rad)	RMSE (rad)	MaxAE (rad)	STD (rad)	90% Percentile Error (rad)
Full Proposed	0.0328	0.0438	0.1346	0.0292	0.0723
Least Square + Sim2Real + KF	0.0581	0.0675	0.1666	0.0344	0.0992
Koopman + RLS	0.0566	0.0756	0.2487	0.0502	0.1229
Offline Koopman	0.0988	0.1402	0.4534	0.0986	0.2408

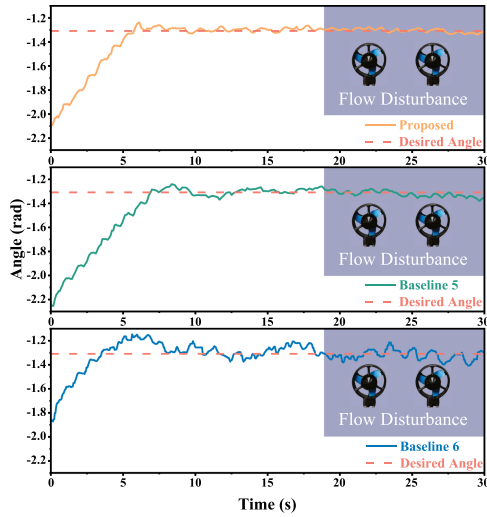


图7. 乌龟仿生机器人在执行方向跟踪任务时航向角的变化。提出方法：采用在线更新的Koopman模型的MPC；基准5：采用离线Koopman模型的MPC；基准6：采用CFD参数模型的MPC。

螺旋桨用于产生可控的流动扰动，从而模拟在现实运行条件下遇到的不确定动力学。为了确保不同动态模型之间的可比性，在一致的控制参数设置下，基于MPC的姿态跟踪比较实验已被执行。

实验结果如图7所示。与CFD参数模型相比，无论是离线Koopman模型还是在线Koopman模型在流动扰动下均表现出更低的超调，并展示了更好的姿态保持性能。其中，所提出的在线Koopman模型进一步提高了跟踪稳定性，表明其在捕捉仿真中未建模动力学方面的有效性。在稳态跟踪阶段，所提出的方法实现了0.0164 rad的均方根误差（RMSE），低于离线Koopman模型的0.0294 rad和CFD参数模型的0.0448 rad。与基线方法相比，所提出方法的跟踪误差降低了63.4%，验证了在线更新机制在提高模型预测精度和扰动抵抗能力方面的有效性。图8展示了实验过程中机器人运动姿态的序列，从视觉上显示了其运动特征。

C. Ablation Study

表 II 中呈现的消融研究结果定量地展示了所提出框架中每个组件的贡献。表 II 中的数据是五次试验的平均值

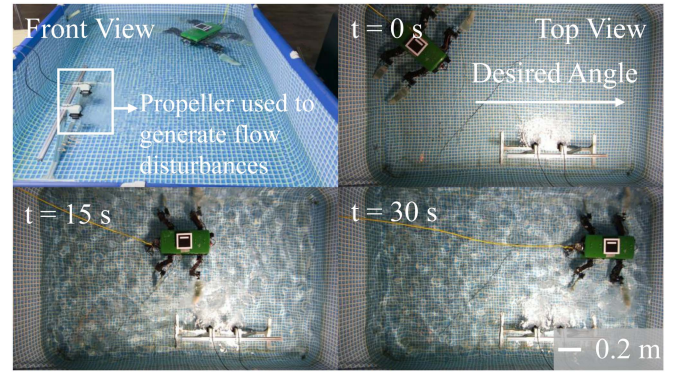


图8. 乌龟灵感机器人在方向追踪任务中的实验快照。

实验运行中，选择的评估指标包括平均绝对误差（MAE）、均方根误差（RMSE）、最大绝对误差（MaxAE）、误差标准差（STD）以及90%百分位误差。首先，在线自适应的必要性通过事实得到证明：使用RLS更新的Koopman模型明显优于离线Koopman模型，这表明仅靠仿真数据无法完全捕捉真实世界的动态。其次，全流程提议方法验证了卡尔曼滤波器作为在线更新机制的优越性，该方法在所有指标上均达到最低误差（MAE 0.0328, RMSE 0.0438），超过了Koopman-RLS变体。最后，通过将全流程提议方法与最小二乘识别模型进行比较，凸显了Koopman提升的重要性：尽管两者都采用Sim2Real预训练和卡尔曼滤波，基于Koopman的架构仍产生显著更低的预测误差（例如，MAE 0.0328 对比 0.0581）。这些比较表明

六、结论

在这封信中，我们提出了一种基于库普曼算子的在线系统识别方法，并将其整合到Sim2Real迁移框架中，该方法已成功应用于受海龟启发的机器人系统的流体动力建模。基于卡尔曼滤波器设计的在线算法能够利用有限的真实实验数据，对离线训练的先验模型进行快速且精确的在线更新，有效捕捉动态环境中系统的时变特性。实验结果表明，所提出的库普曼模型在偏航运动预测和控制任务中均优于传统的参数化CFD模型和最小二乘识别方法，同时保持了强

robustness and tracking accuracy even under disturbances generated by propellers. Moreover, by adopting the Sim2Real transfer strategy, we significantly reduce the reliance on extensive real motion data and enhance the applicability of data driven modeling in scenarios with limited experimental conditions.

Future work will focus on the translation-invariant formulation of the Koopman model that relies only on velocities and error states.

REFERENCES

- [1] K. Melo, T. Horvat, and A. J. Ijspeert, "Animal robots in the african wilderness: Lessons learned and outlook for field robotics," *Sci. Robot.*, vol. 8, no. 85, 2023, Art. no. eadd8662.
- [2] R. Baines et al., "Multi-environment robotic transitions through adaptive morphogenesis," *Nature*, vol. 610, no. 7931, pp. 283–289, 2022.
- [3] R. Wang et al., "Soft robotic fish actuated by bionic muscle with embedded sensing for self-adaptive multiple modes swimming," *IEEE Trans. Robot.*, vol. 41, pp. 1329–1345, 2025.
- [4] T. Salumäe, A. Chemori, and M. Kruusmaa, "Motion control of a hovering biomimetic four-fin underwater robot," *IEEE J. Ocean. Eng.*, vol. 44, no. 1, pp. 54–71, Jan. 2019.
- [5] X. Li, Q. Ren, and J.-X. Xu, "Precise speed tracking control of a robotic fish via iterative learning control," *IEEE Trans. Ind. Electron.*, vol. 63, no. 4, pp. 2221–2228, Apr. 2016.
- [6] Z. Wu, J. Yu, J. Yuan, and M. Tan, "Towards a gliding robotic dolphin: Design, modeling, and experiments," *IEEE/ASME Trans. Mechatron.*, vol. 24, no. 1, pp. 260–270, Feb. 2019.
- [7] R. Wang, S. Wang, Y. Wang, L. Cheng, and M. Tan, "Development and motion control of biomimetic underwater robots: A survey," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 52, no. 2, pp. 833–844, Feb. 2022.
- [8] R. K. Katzschmann, J. DelPreto, R. MacCurdy, and D. Rus, "Exploration of underwater life with an acoustically controlled soft robotic fish," *Sci. Robot.*, vol. 3, no. 16, 2018, Art. no. eaar3449.
- [9] G. Mamakoukas, M. L. Castano, X. Tan, and T. D. Murphey, "Derivative-based Koopman operators for real-time control of robotic systems," *IEEE Trans. Robot.*, vol. 37, no. 6, pp. 2173–2192, Dec. 2021.
- [10] G. Picardi, R. Lovecchio, and M. Calisti, "Towards autonomous area inspection with a bio-inspired underwater legged robot," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2021, pp. 930–935.
- [11] D. Zhang et al., "Development and three-dimensional modeling of a bionic robotic fish with pectoral fins," *Ocean Eng.*, vol. 321, 2025, Art. no. 120393.
- [12] Y. Wang, R. Wang, S. Wang, M. Tan, and J. Yu, "Underwater bioinspired propulsion: From inspection to manipulation," *IEEE Trans. Ind. Electron.*, vol. 67, no. 9, pp. 7629–7638, Sep. 2020.
- [13] W. Wang et al., "Three-dimensional modeling of a fin-actuated robotic fish with multimodal swimming," *IEEE/ASME Trans. Mechatron.*, vol. 23, no. 4, pp. 1641–1652, Aug. 2018.
- [14] A. Liu et al., "An intelligent bionic amphibious turtle robot with visual-tactile fusion for dynamic terrain adaptation," *IEEE Trans. Robot.*, vol. 41, pp. 6345–6363, 2025.
- [15] S. H. Rudy, S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Data-driven discovery of partial differential equations," *Sci. Adv.*, vol. 3, no. 4, 2017, Art. no. e1602614.
- [16] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proc. Nat. Acad. Sci.*, vol. 113, no. 15, pp. 3932–3937, 2016.
- [17] Y. Li and Z. Liu, "Data-driven predictive tracking control of soft robots: A Koopman-MPC framework integrated with variable structure reduced-order extended state observer and kalman filter," *Nonlinear Dyn.*, pp. 1–23, 2025.
- [18] X. Guan, Y. Wang, X. Kang, W. Yao, J. Zhang, and G. Li, "An online system identification algorithm for spherical robot using the koopman theory," *IEEE Robot. Automat. Lett.*, vol. 10, no. 5, pp. 4644–451, May 2025.
- [19] D. Bruder, X. Fu, R. B. Gillespie, C. D. Remy, and R. Vasudevan, "Koopman-based control of a soft continuum manipulator under variable loading conditions," *IEEE Robot. Automat. Lett.*, vol. 6, no. 4, pp. 6852–6859, Oct. 2021.
- [20] C. Ren, H. Jiang, C. Li, W. Sun, and S. Ma, "Koopman-operator-based robust data-driven control for wheeled mobile robots," *IEEE/ASME Trans. Mechatron.*, vol. 28, no. 1, pp. 461–472, Feb. 2023.
- [21] J. Pan, D. Li, J. Wang, P. Zhang, J. Shao, and J. Yu, "Autogeneration of mission-oriented robot controllers using Bayesian-based koopman operator," *IEEE Trans. Robot.*, vol. 40, pp. 903–918, 2024.
- [22] H. Wang, W. Liang, B. Liang, H. Ren, Z. Du, and Y. Wu, "Robust position control of a continuum manipulator based on selective approach and Koopman operator," *IEEE Trans. Ind. Electron.*, vol. 70, no. 12, pp. 12522–12532, Dec. 2023.
- [23] F. Liu, F. Sun, B. Fang, X. Li, S. Sun, and H. Liu, "Hybrid robotic grasping with a soft multimodal gripper and a deep multistage learning scheme," *IEEE Trans. Robot.*, vol. 39, no. 3, pp. 2379–2399, Jun. 2023.
- [24] S. Choi et al., "Learning quadrupedal locomotion on deformable terrain," *Sci. Robot.*, vol. 8, no. 74, 2023, Art. no. eade2256.
- [25] R. Schindeler and K. Hashtudi-Zaad, "Online identification of environment hunt-crossley models using polynomial linearization," *IEEE Trans. Robot.*, vol. 34, no. 2, pp. 447–458, Apr. 2018.
- [26] L. Hong, X. Wang, D.-S. Zhang, M. Zhao, and H. Xu, "Vision-based underwater inspection with portable autonomous underwater vehicle: Development, control, and evaluation," *IEEE Trans. Intell. Veh.*, vol. 9, no. 1, pp. 2197–2209, Jan. 2024.
- [27] J. A. Andersson, J. Gillis, G. Horn, J. B. Rawlings, and M. Diehl, "CasADi: A software framework for nonlinear optimization and optimal control," *Math. Program. Computation*, vol. 11, no. 1, pp. 1–36, 2019.
- [28] A. Liu, Y. Xue, H. Qin, and Z. Zhu, "Physics-informed identification of marine vehicle dynamics using hydrodynamic dictionary library-inspired adaptive regression," *Ocean Eng.*, vol. 296, 2024, Art. no. 117013.

即使在螺旋桨产生的干扰下，也能保持稳健性和跟踪精度。此外，通过采用Sim2Real迁移策略，我们显著减少了对大量真实运动数据的依赖，并增强了数据驱动建模在实验条件有限的场景中的适用性。

未来的工作将集中于依赖于速度和误差状态的、平移不变的库普曼模型公式化。

参考文献

- [1] K. Melo, T. Horvat, 和 A. J. Ijspeert, “非洲荒野中的动物机器人：经验教训与野外机器人展望,” *Sci. Robot.*, 第8卷, 第85期, 2023年, 文章编号 eadd8662. [2] R. Baines 等, “通过自适应形态发生的多环境机器人过渡,” *Nature*, 第610卷, 第7931期, 页283–289, 2022年. [3] R. Wang 等, “由嵌入感知的仿生肌肉驱动的软体机器人鱼, 实现自适应多模式游泳,” *IEEE Trans. Robot.*, 第41卷, 页1329–1345, 2025年. [4] T. Salumäe, A. Chemori, 和 M. Kruusmaa, “悬浮仿生四鳍水下机器人的运动控制,” *IEEE J. Ocean. Eng.*, 第44卷, 第1期, 页54–71, 2019年1月. [5] X. Li, Q. Ren, 和 J.-X. Xu, “通过迭代学习控制实现机器人鱼的精确速度跟踪,” *IEEE Trans. Ind. Electron.*, 第63卷, 第4期, 页2221–2228, 2016年4月. [6] Z. Wu, J. Yu, J. Yuan, 和 M. Tan, “面向滑翔的机器人海豚：设计、建模与实验,” *IEEE/ASME Trans. Mechatron.*, 第24卷, 第1期, 页260–270, 2019年2月. [7] R. Wang, S. Wang, Y. Wang, L. Che [14] A. Liu 等, “一种用于动态地形适应的视觉-触觉融合智能仿生两栖乌龟机器人,” *IEEE Trans. Robot.*, 第41卷, 页6345–6363, 2025. [15] S. H. Rudy, S. L. Brunton, J. L. Proctor, 和 J. N. Kutz, “基于数据发现偏微分方程,” *Sci. Adv.*, 第3卷, 第4期, 2017, 文章号 e1602614. [16] S. L. Brunton, J. L. Proctor, 和 J. N. Kutz, “通过非线性动力系统稀疏识别从数据中发现控制方程,” *Proc. Nat. Acad. Sci.*, 第113卷, 第15期, 页3932–3937, 2016. [17] Y. Li 和 Z. Liu, “软体机器人基于数据的预测追踪控制：一种结合可变结构低阶扩展状态观测器和卡尔曼滤波器的 Koopman-MPC 框架,” *Nonlinear Dyn.*, 页1–23, 2025. [18] X. Guan, Y. Wang, X. Kang, W. Yao, J. Zhang, 和 G. Li, “用于球形机器人的在线系统识别算法, 基于 Koopman 理论,” *IEEE Robot. Automat. Lett.*, 第10卷, 第5期, 页4644–451, 2025年5月. [19] D. Bruder, X. Fu, R. B. Gillespie, C. D. Remy